Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/04/2005
PI Name:	Head, Nathan Ph.D.		
Project Title:	Genetic Regulation of Pseudomonas aeruginosa Biofilms in Space		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	25704	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	GSRP
Start Date:	07/01/2004	**End Date:**	06/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:			
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Performance Goal Text:			
Task Description:	Pseudomonas aeruginosa readily forms biofilms, or exopolysaccharide-surrounded bacterial communities, during space travel inside vehicles. These bacterial growths may cause multi-focal low grade pneumonia to the inhabitants of these vehicles due to the biofilm being surrounded by the protective polysaccharide coating. I have identified four novel loci on the P. aeruginosa PAO1 genome that regulate overproduction of an exopolysaccharide called alginate needed for the formation of mature biofilms. In this study, a gene inactivation system will be developed and used to inactivate these genes on the PAO1 chromosome to show phenotypic relevance and involvement of these genes in alginate regulation. To identify novel genes currently present within the P. aeruginosa PAO1 genome that regulate the production of alginate. Identification of novel genes that control the production of alginate can help shed light on the alginate regulatory pathway, which currently, is thought to be fully understood.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Adequately discerning the full scope of the alginate regulation pathway will allow for proper advances in inhibition of P. aeruginosa biofilm formation on abiotic surfaces. By identifying and utilizing some of these novel genes found to regulate alginate production, proper treatment to these surfaces can aid in the suppression of these growths, and potentially translate to biologically safer microgravity conditions for inhabitants of space vehicles.
Task Progress:	Since the inception of this project, further studies have been performed on one of the four cosmids shown to reverse the overproduction of an exopolysaccharide called alginate, excreted from the bacterium Pseudomonas aeruginosa mutant strain PAO579. Genetic analysis of this genomic cosmid (MTP87) that effects the production of alginate was performed by randomly inactivating the cosmid and determining if the effect previously seen is negated. A portion of this cosmid was shown to play a role in the regulation of alginate production. cysP, a gene that creates a sulfate-binding protein of the ABC transporter, appears to be responsible for suppressing the overproduction of alginate in P. aeruginosa. However, after constructing a gene knockout of this gene (explained in the original proposal) in the parent strain, PAO1, this effect was not clearly seen. Further observation showed a very interesting factor; when incubating these cysP knockout bacteria at 37 degrees Celsius, no change in alginate production was seen. However, after removing the samples to room temperature (about 25 degrees Celsius) over a period of a few days, these bacteria began to produce more alginate. This effect seems to be temperature dependent. Further biochemical analysis of this effect is being determined. A similar situation was found in a different mutant strain PAO568. A cosmid previously identified to suppress alginate production in this mutant (MTP93) was subjected to random inactivation. A portion of this cosmid was found to regulate alginate production, an open reading frame and proposed gene named PA1592. Gene knockouts are currently being performed to confirm this finding. Another method, not previously proposed, involves random inactivation of the parent strain PAO1 genome and screening this for mutants that overproduce alginate, in the effort of identifying novel areas of the PAO1 genome that control alginate production. This procedure has led to the identification of many different novel locations on the PAO1 genome that are currently being confirmed.
Bibliography Type:	Description: (Last Updated:)